Assignment No: 2

Title: Develop a webpage using semantic HTML tags.

Aim: To design and develop a structured, modern webpage from scratch using HTML5 semantic elements.

Objectives:

1. To learn the use of basic semantic tags in HTML documents.
2. To understand and manage webpage layout elements.
3. To apply semantic landmarks to a webpage.
4. To understand the role of HTML5 semantic elements in webpage structure.
5. To create a meaningful and accessible webpage layout using semantic tags.

Hardware Used:

- Computer or Laptop
- Keyboard
- Mouse
- Minimum 4 GB RAM

Software Used:

- Visual Studio Code (VS Code)
- Web Browser (Google Chrome, Microsoft Edge, or Firefox)
- Operating System: Windows / Linux / macOS

Theory:

1. Introduction to Semantic HTML

Semantic HTML refers to the practice of using HTML tags that clearly describe the meaning, role and context of the content inside them, rather than focusing only on how the content should look.

Semantic elements make the structure of a webpage easier to understand for developers, browsers, search engines and assistive technologies.

2. Important Semantic HTML Tags

Tag	Purpose
<header>	Introductory content for a page or a section; title, logo, intro.
<nav>	The main navigation links of a webpage.
<main>	The primary content of the document; only one per page.
<section>	Groups related content under a common topic or heading.

<article>	Self-contained content that can be understood independently.
<aside>	Secondary content such as a sidebar or quick tips.
<figure> / <figcaption>	An illustration together with its caption.
<footer>	Footer information: copyright, contact details, useful links.

3. Benefits of Semantic HTML

- Improves webpage structure and readability.
- Helps screen readers and assistive technologies understand the page.
- Makes website content easier for search engines to interpret.
- Improves code organization and maintainability.
- Helps developers create meaningful and accessible websites.

4. Semantic HTML Page Structure

A typical webpage is organized as:

```
<header> ... site title and navigation ... </header>
<main>
  <section>
    <article> ... individual article ... </article>
  </section>
  <aside> ... related information ... </aside>
</main>
<footer> ... copyright and contact ... </footer>
```

This structure provides a clear layout and improves the understanding of webpage content by browsers and assistive technology.

Course Project Implementation:

Course-Project Mapping: Partially available. The LipiSutra project does use semantic landmarks, but only three of them and only inside JSX: src/App.jsx renders <header> (line 70) and <main> (line 86), and src/components/Navbar.jsx renders <nav> (line 11). There is no <section>, <article>, <aside>, <figure> or <footer> anywhere in the project, so those three files alone do not satisfy the assignment. Submit the complete semantic page given below, and cite App.jsx and Navbar.jsx as the project evidence of the same landmarks used in production.

Semantic element	In LipiSutra?	File
<header>	Yes	src/App.jsx (line 70)
<nav>	Yes	src/components/Navbar.jsx (line 11)
<main>	Yes	src/App.jsx (line 86)
<section>, <article>, <aside>, <footer>	No	supplied below

Program:

File: assignment02_semantic.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>LipiSutra - Semantic Layout</title>
  <style>
    body   { font-family: Georgia, serif; margin: 0; color: #1a0800; background:
#f5ede0; }
    header { background: #2a1800; color: #D4A017; padding: 20px 32px; }
    nav ul { list-style: none; display: flex; gap: 24px; margin: 0; padding: 12px
32px;
             background: #3a2410; }
    nav a  { color: #f5e6c8; text-decoration: none; font-size: 14px; }
    main   { padding: 24px 32px; }
    article{ background: #fff; border-left: 4px solid #8B4513; padding: 16px; margin-
bottom: 20px; }
    aside  { background: #ede0cc; border: 1px dashed #8B4513; padding: 16px; }
    footer { background: #2a1800; color: #c9a96e; padding: 16px 32px; font-size: 13px;
}
  </style>
</head>
<body>

  <!-- HEADER : introductory content of the page -->
  <header>
    <h1>LipiSutra</h1>
    <p>Reading what history almost forgot.</p>
  </header>

  <!-- NAV : primary navigation landmark -->
  <nav>
    <ul>
      <li><a href="#scanner">AI Scanner</a></li>
      <li><a href="#archive">Archives</a></li>
      <li><a href="#roles">Roles</a></li>
      <li><a href="#about">About</a></li>
    </ul>
  </nav>

  <!-- MAIN : the single primary content region -->
  <main>

    <!-- SECTION : groups related articles under one topic -->
    <section id="scanner">
      <h2>How the Decoder Works</h2>

      <!-- ARTICLE : self-contained, independently meaningful -->
      <article>
        <h3>Agent 1 - OCR and Transliteration</h3>
        <p>
          The uploaded artifact is passed to a vision model that detects the script
          and produces a literal transliteration of every clearly visible glyph.
        </p>
      </article>

      <article>
        <h3>Agent 2 - Gap Filling and Translation</h3>
        <p>
          Torn and faded regions are reconstructed from context, then the restored
          text is rendered into modern Marathi and summarised in English.
        </p>
      </article>
    </section>

    <section id="roles">
      <h2>Platform Roles</h2>
```

```
      <!-- FIGURE / FIGCAPTION : illustration with its caption -->
      <figure>
        <img src="hero.png" alt="Historian reviewing a decoded manuscript"
width="380">
        <figcaption>Figure 1: The historian verification workspace.</figcaption>
      </figure>

      <article>
        <h3>Public, Historian and Museum Admin</h3>
        <p>
          A reader uploads and reads. A historian corrects and verifies. An
          administrator monitors the digitization statistics of the collection.
        </p>
      </article>
    </section>

    <!-- ASIDE : secondary content, related but not essential -->
    <aside id="about">
      <h3>Did you know?</h3>
      <p>
        The Modi script was the administrative hand of the Maratha empire and is
        today read fluently by only a few hundred people.
      </p>
    </aside>

  </main>

  <!-- FOOTER : closing information for the document -->
  <footer>
    <p>&copy; 2026 LipiSutra &middot; Department of Computer Engineering, KBTCOE
Nashik.</p>
    <address>Contact: team@lipisutra.example</address>
  </footer>

</body>
</html>
```

Project evidence - src/App.jsx (semantic landmarks in production)

```
<header className="absolute top-4 right-4 z-[100]">
  <button onClick={toggleTheme} ...>
    {themeMode === "DARK" ? "Dark" : "Light"}
  </button>
</header>

<Navbar currentView={view} onViewChange={setView} theme={theme} />

<main className="flex-1 overflow-x-hidden relative z-10 w-full">
  {view === "public"    && <PublicView    theme={theme} />}
  {view === "historian" && <HistorianView theme={theme} />}
  {view === "museum"    && <MuseumView    theme={theme} />}
</main>
```

Output:

- The page renders as four clearly separated bands: a dark header, a navigation strip, the main content region, and a footer.
- Within <main>, two <section> blocks group the pipeline articles and the role articles.
- The <aside> renders as a bordered side note, visually distinct from the articles.
- Opening the browser accessibility tree (DevTools > Accessibility) lists the landmarks banner, navigation, main, complementary and contentinfo without any ARIA attributes being written by hand.

Conclusion:

In this assignment, we designed and developed a structured webpage using HTML5 semantic elements. We learned the purpose of <header>, <nav>, <main>, <section>, <article>, <aside>, <figure> and <footer> tags, and we verified in the browser accessibility tree that these tags automatically expose landmark roles. We also located the same landmarks inside the LipiSutra course project, where <header>, <nav> and <main> are produced from JSX.

References:

6. Mozilla Developer Network (MDN) Web Docs - HTML Elements, https://developer.mozilla.org/en-US/docs/Web/HTML/Element
7. W3Schools - HTML Semantic Elements, https://www.w3schools.com/html/html5_semantic_elements.asp
8. HTML Living Standard - WHATWG, https://html.spec.whatwg.org/